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Department of Artificial Intelligence
Intelligent Multi-Agent and Expert Systems (AI401), Semester VII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_9_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 9

Knowledge Acquisition — Elicited Rules and Rules Induced by Machine Learning

Aim

To audit a rule base obtained from an expert by interview, finding the cases it leaves uncovered and the cases where its rules contradict one another, and then to induce a rule base for the same domain from a set of solved cases by decision-tree learning and compare the two.

Theory

The knowledge base of Experiment 2 had to come from somewhere, and getting it there is the hardest part of building an expert system. Elicitation is the direct route: the knowledge engineer interviews the expert and writes down production rules. It fails in characteristic ways rather than at random. An expert states the cases that come to mind and omits the ones that do not, leaving combinations of attribute values that no rule matches; and rules given at different moments overlap on cases neither was written for, so two of them fire with different conclusions. Neither defect is visible in the rules themselves — both appear only when the rule base is run against the space of possible cases, which is what makes the audit mechanical.

Machine learning offers the indirect route: instead of asking the expert for rules, collect cases the expert has already solved and induce the rules from them. A decision-tree learner chooses at each step the attribute whose value most reduces the entropy of the class labels, splits on it and recurses; every path from root to leaf is then a production rule, and because the tree partitions the attribute space, the resulting rule base is complete and conflict-free by construction. What it buys in consistency it pays for in dependence on the cases: the rules can only be as good as the sample, and a mislabelled case is learned as faithfully as a correct one.

Algorithm

Algorithm 1 — Auditing an elicited rule base

1. Fix the attributes of the domain with their possible values and enumerate the complete space of cases they define. Write down the rule base as obtained from the expert, each rule being a set of attribute tests and a conclusion.
2. For every case in the space, collect the rules that fire. Report the cases for which no rule fires, the cases for which two or more rules fire with different conclusions together with the rules involved, and the accuracy of the rule base when the first matching rule is allowed to decide.

Algorithm 2 — Inducing a rule base from solved cases

1. Assemble a case base of solved examples, each a complete assignment of the attributes together with the conclusion the expert reached. Implement the entropy of a set of cases and the information gain of splitting it on an attribute.
2. Build the tree recursively: stop with a leaf when every case carries the same conclusion or no attribute remains, otherwise split on the attribute of highest gain and recurse on each value. Print the tree, convert every root-to-leaf path into a production rule, and report the number of rules, the accuracy over the whole case space and how many of the elicited rules the induced tree agrees with.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the program below, including the elicited rule base, the lists of uncovered and conflicting cases with the rules responsible, the induced tree, the rules read off from it and the accuracy of both rule bases over the complete case space.

Observations

Comment on which defect of the elicited base is more dangerous in use, a case no rule covers or a case two rules answer differently, on why ordering the rules and taking the first match hides the second defect without removing it, on whether the rules the induced tree refuses to confirm are the same ones the audit flagged, and on what the expert still supplies once the rules themselves are being learned from cases.

Exercises

1. Corrupt a fraction of the conclusions in the case base, as an expert labelling inconsistently would, and report the number of leaves in the induced tree and its accuracy at corruption levels of 0, 5, 10 and 20 per cent, averaged over several case bases. Then re-run the induction with a rule that refuses to split a set smaller than a stated size, and report whether it helps at each level.
2. Vary the size of the case base over 8, 16, 24, 40 and 80 cases, averaging over several draws, and report the number of leaves, the accuracy over the whole case space and how many of the elicited rules the induced tree agrees with. State the number of cases at which induction overtakes the elicited rule base, and explain why the count of agreed rules does not rise with the size of the case base.